Young Jin Lee

Email: youngjinlee@gatech.edu | **LinkedIn** | **GoogleScholar** | **Mobile:** (+1)404-940-4307

Summary

Interdisciplinary researcher specializing in **thermal management of high-power-density electronics**, and **electrical-thermal co-design of advanced semiconductor packaging**. Doctoral research focused on silicon-embedded liquid cooling of ultra-high power density (> 1 kW/cm^2) electronics using manifold microchannel coolers. Highly experienced in hands-on experiments, including single/two-phase flow loop testbed and cleanroom fabrication of semiconductor packaging, along with expertise in modeling and optimization using computational fluid dynamics (CFD) simulation. **Author of eight publications** in peer-reviewed journals and conferences, including **seven lead-author papers**.

Research interests

Silicon-embedded liquid cooling of high-power-density electronics

Electrical-thermal co-design for advanced electronics packaging

Thermal management of three-dimensional integrated circuits (3D-ICs)

Professional Experience

Georgia Institute of Technology Atlanta, GA, United States

Postdoctoral Fellow (On-site, 100% appointment) *Apr. 2025 – Present*

- Postdoc at Integrated 3D Systems (I3DS) group in the School of Electrical and Computer Engineering (Advisor: Prof. Muhannad S. Bakir)
- Support research and development in the areas of advanced semiconductor packaging and heterogeneous integration, including advanced cooling, 2.5D and 3D IC technologies, and new interconnects architecture

Semiconductor Research Corporation (SRC) Atlanta, GA, United States

Research Scholar *Sep. 2025 – Present*

- Research scholar in SRC/DARPA JUMP 2.0 CogniSense (Center on Cognitive Multispectral Sensors) program
- Develop an electrical-thermal co-functional copper bridge monolithically integrated into 3D RFIC-antenna packages (see Research Projects [R6], and Honors & Awards)

Education

Korea Advanced Institute of Science and Technology (KAIST) Daejeon, South Korea

Ph.D Mechanical Engineering *Mar. 2020 – Feb. 2025*

- Dissertation Title: Embedded liquid cooling with manifold microchannel for ultra-high heat flux electronics (Advisor: Prof. Sung Jin Kim)

M.S. Mechanical Engineering *Mar. 2018 – Feb. 2020*

- Thesis Title: Optimal thermal design of pin-fin heat sinks with variable fin density in natural convection (Advisor: Prof. Sung Jin Kim)

B.S. Mechanical Engineering *Mar. 2013 – Feb. 2018*

- Honors: First prize in *International Cooperative Design* (Advisor: Prof. Jeong-Hoi Koo at Miami University) (2015), First prize in *Multidisciplinary Capstone Design* (2017)

Journal Publications

[1]: *Co-first authorship*

J6 E. Chung[1], **Y.J. Lee**[1], H. Lee, M.S. Bakir, 3D Heterogeneous Integration with Microfluidic Cooling: Addressing Thermal Management and Interconnect Co-design Challenges, ***IEEE Transactions on Components, Packaging and Manufacturing Technology***, under review, **2026**

J5 **Y.J. Lee**[1], CH Hwang[1], H. Lee[1], I. Lee, S.J. Kim, Highly energy-efficient manifold microchannel for cooling electronics with a coefficient of performance over 100,000, ***Energy Conversion and Management*** 121422, **2026**

Media coverage: "Energy-Efficient Liquid Cooling for Advanced Semiconductor Packaging", ***Semiconductor Engineering (US)***, *Apr. 2026.* [Link]

"New cost-effective internal cooling solution solves semiconductor chip overheating", ***Interesting Engineering (US)***, *June. 2026.* [Link]

"KAIST breaks a major AI bottleneck with liquid cooling technology 10 times more efficient than the previous record", ***EurekaAlert! by AAAS (American Association for the Advancement of Science)***, *June. 2026.* [Link]

J4 H. Lee[1], **Y.J. Lee**[1], S.J. Kim, One-dimensional model of manifold microchannels for embedded cooling: Prediction of thermal performance and flow non-uniformity, ***International Communications in Heat and Mass Transfer*** 107498, **2024**

J3 **Y.J. Lee**, S.J. Kim, Experimental investigation on thermal-hydraulic performance of manifold microchannel with pin-fins for ultra-high heat flux cooling, ***International Journal of Heat and Mass Transfer*** 125336, **2024**

J2 J. Kim, **Y.J. Lee**, J.-Y. Kim, J. Yoon, H. Yoon, J.-B. You, S.J. Kim, H. Kurt, and H.-H. Park, High-efficiency thermo-optic control of longitudinal radiation angle in silicon-based optical phased array by leveraging backside air trench structure, ***Optics Letters*** 524469, **2024**

J1 **Y.J. Lee**, S.J. Kim, Thermal optimization of the pin-fin heat sink with variable fin density cooled by natural convection, ***Applied Thermal Engineering*** 116692, **2021**

Peer-reviewed Conference publications

C3 W. Zeng, **Y.J. Lee**, M.S. Bakir, Electrical-Thermal Co-Design and Fabrication of an Interlayer Copper Bridge for 3D mmWave Systems, *International Interconnect Technology Conference (**IITC**)*, **2026**

C3 **Y.J. Lee**, H. Lee, S.J. Kim, Analytical prediction of thermal performance and flow non-uniformity of manifold microchannel heat sinks, *23rd IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (**ITherm**)*, **2024**

C2 **Y.J. Lee**, S.J. Kim, Experimental investigation on thermal-hydraulic performance of 3-D manifold microchannel (MMC) for high heat flux electronics, *17th International Heat Transfer Conference (**IHTC**)*, **2023**

C1 **Y.J. Lee**, S.J. Kim, Compact modeling of the pin-fin heat sink with variable fin density cooled by natural convection, *The Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (**ITherm**)*, **2021**

Conference oral presentation

O7 **Y.J. Lee**, W. Zeng, M.S. Bakir, Monolithic interlayer copper heat sink for heterogeneously integrated 3D RFIC-Antenna Package, *SRC/DARPA JUMP 2.0 CogniSense Annual Review*, **2025**

O6 **Y.J. Lee**, CH Hwang, H. Lee, I. Lee, and S.J. Kim, Optimization and experimental investigation on silicon manifold microchannel for embedded cooling, *3rd Pacific Rim Thermal Engineering Conference (**PRTEC**)*, **2024**

O5 **Y.J. Lee**, S.J. Kim, Thermal-hydraulic performance of embedded manifold microchannel with pin-fins: Experimental study, *Korean Society Mechanical Engineering Thermal Engineering Division (**KSME-TED**)*, **2024**

O4 **Y.J. Lee**, S.J. Kim, 3-D Manifold Microchannel (MMC) with Pin-Fins and Plate-Fins for Silicon-Based Embedded Cooling of Ultra-High Heat Flux Electronics, *International Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Microsystems (**InterPACK**)*, **2022**

O3 **Y.J. Lee**, Y.J. Lee, S.J. Kim, Loop heat pipe heat exchanger (LHP-HX) for recovering liquefied natural gas (LNG) cold energy, *The 2nd Asian Conference on Thermal Sciences (**ACTS**)*, **2021**

O2 **Y.J. Lee**, S.J. Kim, Compact modeling and optimization of the pin-fin heat sink with variable fin density cooled by natural convection, *Korean Society Mechanical Engineering Thermal Engineering Division (**KSME-TED**)*, **2021**

O1 **Y.J. Lee**, S.J. Kim, Optimal pin-fin heat sinks with variable fin density cooled by natural convection, *Korean Society Mechanical Engineering Thermal Engineering Division (**KSME-TED**)*, **2020**

Patents

P6 Heat dissipation device (3-D manifold multiporous heat sink for high-heat flux electronics cooling), Korea(KR), Patent Registration Number KR102552852B1, 2023

P5 Heat dissipation device (Manifold microchannel heat sink with pin-fins), Korea(KR), Patent Registration Number KR102841128B1, 2023

P4 Automatic control device for foot rests (Outcome of *Multidisciplinary Capstone Design* (CD401)), Korea(KR), Patent Registration Number KR102174705B1, 2020

P3 Wheelchair brake system (Outcome of *Multidisciplinary Capstone Design* (CD401)), Korea(KR), Patent Registration Number KR101956000B1, 2019

P2 Safety system of cross road walk and method for transmitting and receiving data thereof, Korea(KR), Patent Registration Number KR101956000B1, 2013

P1 Method for producing a heavy metal adsorbent with enhanced adsorptive power using sunflower stem, Korea(KR), Patent Registration Number KR101282042B1, 2013

Research Projects

[R6] **Electrical-thermal co-design of a heterogeneous integration platform for 3D RFIC-antenna packages** Sep. 2025 – Present

Sponsored by Semiconductor Research Corporation (SRC) *PI: Prof. Muhannad S. Bakir*

- JUMP 2.0 CogniSense Program (Thrust 4: Heterogeneous System Integration)
- Proposed an electrical-thermal co-functional platform for 3D RFIC-antenna packages comprising (1) monolithic copper interlayer serving as electrical ground plane and heat spreader, and (2) copper pillar interconnects for ultra-low insertion loss.
- Fabricated and assembled a test vehicle comprising glass tier with antenna and through-glass-via (TGV), and silicon RFIC tier bonded by Cu-Cu thermocompression bonding.
- Conducted electro-thermal simulations using ANSYS. From an electrical perspective, achieved interconnect insertion loss below 0.5 dB at 170 GHz. From a thermal perspective, demonstrated that a 30 µm copper interlayer acts as an effective heat spreader, maintaining junction temperatures equivalent to external bulky copper heat sinks (∼cm scale) while reducing package volume by over 10×.

[R5] **Manifold microchannel cooling system for ultra-high heat flux applications** Mar. 2021 – Feb. 2025

Sponsored by National Research Foundation of Korea (NRF) *PI: Prof. Sung Jin Kim*

- Developed a thermal-hydraulic model to predict the thermal resistance and flow non-uniformity of a manifold microchannel heat sink (see reference [J4])
- Developed a MEMS fabrication process for silicon-based/glass-based embedded manifold microchannel heat sinks
- Experimentally demonstrated embedded cooling for a heat flux of 1.2 kW/cm^2 without phase-change heat transfer, with a pressure drop of less than 5 kPa at a junction temperature of 110°C (see reference [J3])

[R4] **Design & electric/thermal/material analysis of 3-D PIM-HBM architecture** Mar. 2023 – Feb. 2025

Sponsored by National Research Foundation of Korea (NRF) *PI: Prof. Sung Jin Kim*

- Developed a compact thermal model of PIM heterogeneous packages for thermal reliability verification
- Developed through-silicon via (TSV) embedded manifold microchannel and co-analyzed electrical and thermal performances

[R3] **Electro-thermal co-design of silicon-based optical phased arrays with tunable radiator for energy-efficient beam steering** Mar. 2022 – Feb. 2025

Sponsored by National Research Foundation of Korea (NRF) *PI: Prof. Hamza Kurt*

- Developed an energy-efficient thermo-optic control for longitudinal beam steering in silicon-based optical phased arrays (OPAs) with tunable radiators (in collaboration with Prof. H. Kurt and Prof. H.-H. Park (KAIST)) (see reference [J2])
- Proposed and optimized the backside air trench structure, successfully doubling the energy efficiency of the OPAs compared to the benchmark

[R2] **Loop heat pipe heat exchanger for waste heat recovery of LNG re-gasification process** May 2021 – Oct. 2021

Sponsored by HD Korea Shipbuilding and Offshore Engineering Co., Ltd. *PI: Prof. Sung Jin Kim*

- Proposed a loop heat pipe heat exchanger (LHP-HX) to recover cold energy from liquefied natural gas (LNG) re-gasification process, as an alternative to the conventional pump-assisted active system
- Determined the design parameters of a 500 kW-scale LHP-HX using epsilon-NTU method (presented the results at *The 2nd Asian Conference on Thermal Sciences (ACTS)* — refer to [O3])

[R1] **Weight reduction of 5G MMU station via advanced thermal management** Oct. 2019 – Oct. 2020

Sponsored by Network Business Division, SAMSUNG Electronics *PI: Prof. Sung Jin Kim*

- Proposed and optimized a pin-fin heat sink with a variable fin population, achieving a 30% weight reduction while maintaining the thermal performance (see reference [J1])
- Embedded a passive cooling device—a pulsating heat pipe—into the base substrate, and experimentally demonstrated an effective thermal conductivity of 4000 W/m·K and achieved a 24% weight reduction of the base substrate

Honors & Awards

Best Poster Award (SRC/DARPA JUMP 2.0 CogniSense Annual Review) Oct. 2025

Issued by Semiconductor Research Corporation (SRC)

An honor voted on by industry experts from companies such as Intel, Raytheon, IBM, and Micron, and awarded to fewer than 5% of researchers participating the program (5 out of ∼120).

National Excellence Scholarship Mar. 2020 – Feb. 2025

Full-Tuition Scholarship for Doctoral Program, Ministry of Education, South Korea

KAIST Scholarship for Academic Excellence Mar. 2018 – Feb. 2020

Full-Tuition Scholarship for Master Program, KAIST

National Excellence Scholarship Mar. 2013 – Feb. 2018

Full-Tuition Scholarship for Undergraduate Program, Ministry of Education, South Korea

Professional Activities

Invited reviewer refer to ORCID record

- Completed **44 peer reviews** across 7 international journals

Membership

- Member - Institute of Electrical and Electronics Engineers (IEEE) (link)
- Member - IEEE Electronics Packaging Society (link)

Teaching

Served as a teaching assistant for the following courses, grading exams and quizzes. Held office hours to address student questions and actively supported students in understanding the course material.

[TA] Convective Heat Transfer (ME611)	2023
[TA] Thermodynamics (ME211)	2023
[TA] Heat Transfer (ME321)	2022
[TA] Phase Change Heat Transfer (ME514)	2022
[TA] Fluid Mechanics (ME221)	2020, 2021
[TA] Mathematical Methods in Mechanical Engineering (ME500)	2020

Skills

Experiment

Construction of single/two-phase flow loop testbed, MEMS fabrication techniques (etching, anodic bonding, direct fusion bonding, sputtering etc.), IR thermometry, Material characterization (FE-SEM, TEM, EDS, XRM, XPS, FT-IR etc.)

Numerical Modeling and Optimization

Computational fluid dynamics (ANSYS Fluent, COMSOL Multiphysics), Reduced-order modeling (ROM), Data-driven optimization, Surrogate-model-based optimization (multi-fidelity Kriging method etc.)

References

Sung Jin Kim
Professor
Mechanical Engineering, KAIST
Phone: +82 42-350-3043
Email: sungjinkim@kaist.ac.kr

Hamza Kurt
Professor
Electrical Engineering, KAIST
Phone: +82 42-350-7465
Email: hamzakurt@kaist.ac.kr

Je-Young Chang
Principal Engineer
Intel (USA)
Phone: +1 (480) 678-1019
Email: je-young.chang@intel.com

Muhannad S. Bakir
Professor
Electrical and Computer Engineering, Georgia Institute of Technology
Phone: +1 (404) 385-6276
Email: mbakir@ece.gatech.edu

Kyung-sik (Jack) Moon
Senior Research Scientist
3D Packaging Research Center, Georgia Institute of Technology
Phone: +1 (404) 894-8465
Email: jack.moon@gatech.edu

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